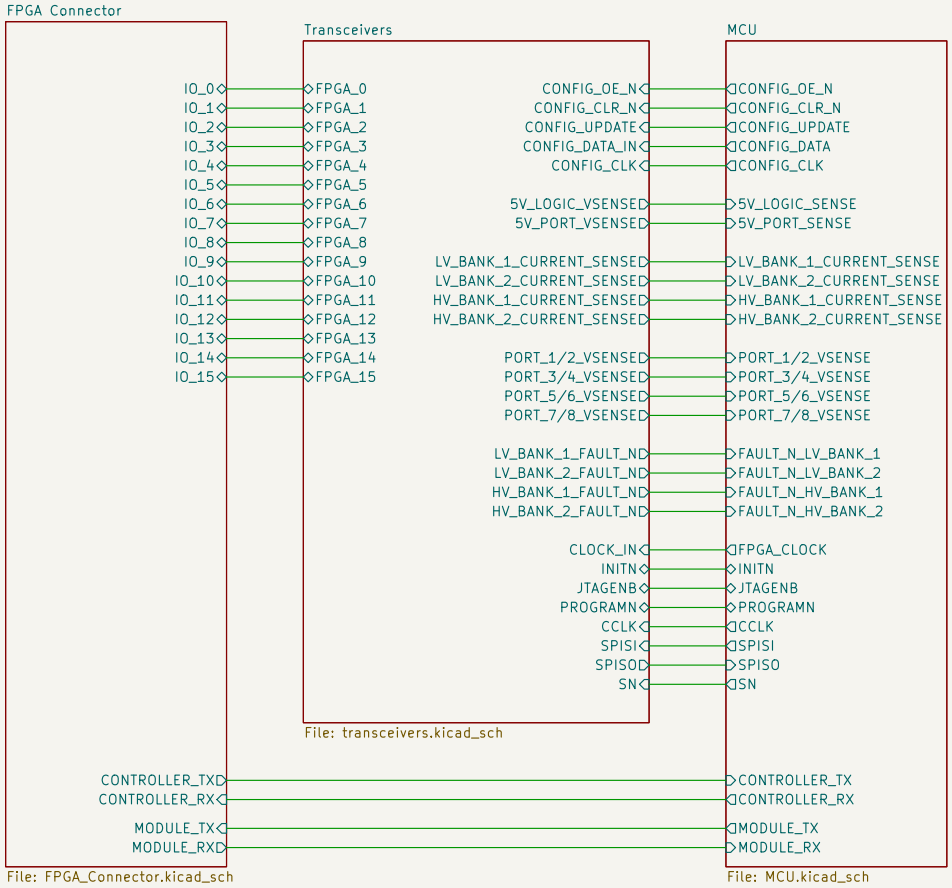
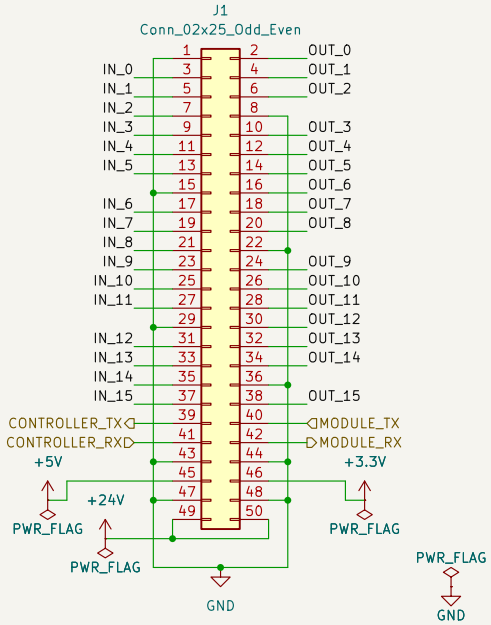
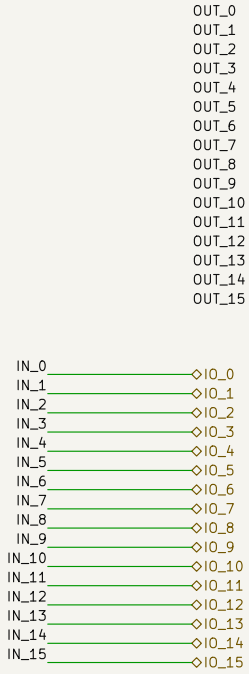


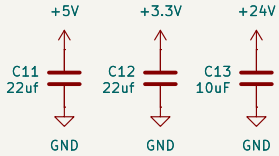
Sheet: /
File: Serial Module.kicad_sch

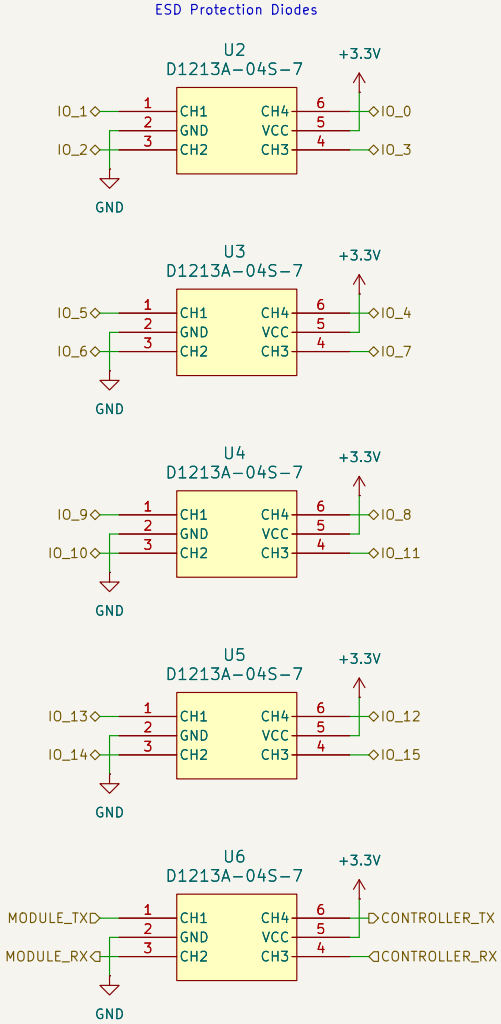
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/20





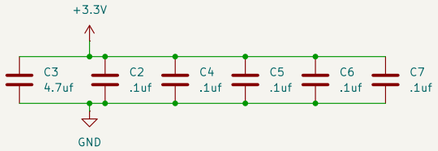


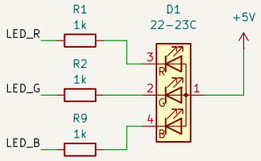


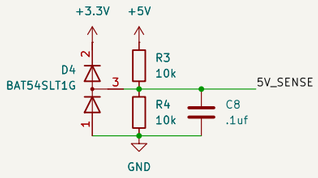
Sheet: /FPGA Connector/
File: FPGA_Connector.kicad_sch

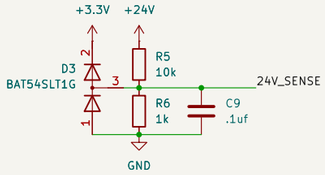
Title:

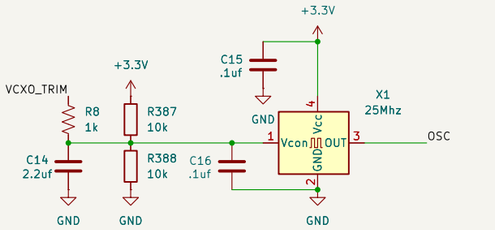
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 2/20

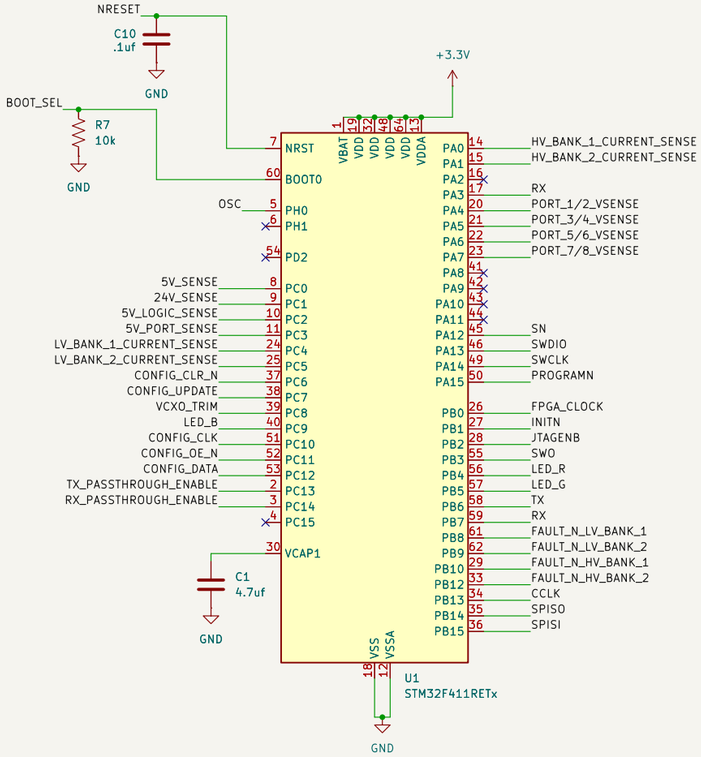


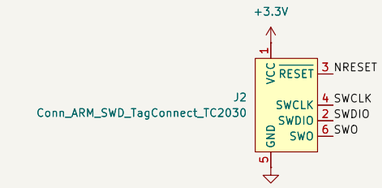


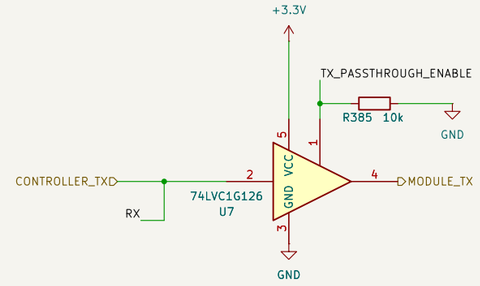




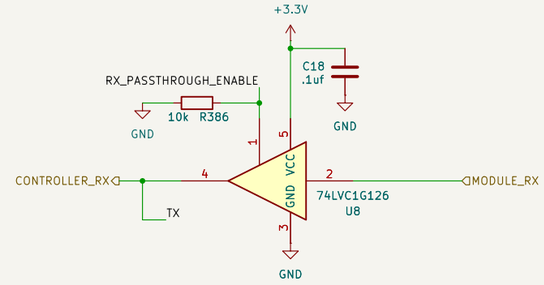












Sheet: /MCU/	
File: MCU.kicad_sch	
Title:	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Id: 3/20

FPGA MUX

PROGRAMN
JTAGENB
INITN
CLOCK_IND
CCLK
SPISI
SPISO
SND

FPGA_0
FPGA_1
FPGA_2
FPGA_3
FPGA_4
FPGA_5
FPGA_6
FPGA_7
FPGA_8
FPGA_9
FPGA_10
FPGA_11
FPGA_12
FPGA_13
FPGA_14
FPGA_15

FPGA_PORT_1_PIN_0
FPGA_PORT_1_PIN_1
FPGA_PORT_1_PIN_2
FPGA_PORT_1_PIN_3
FPGA_PORT_2_PIN_0
FPGA_PORT_2_PIN_1
FPGA_PORT_2_PIN_2
FPGA_PORT_2_PIN_3
FPGA_PORT_3_PIN_0
FPGA_PORT_3_PIN_1
FPGA_PORT_3_PIN_2
FPGA_PORT_3_PIN_3
FPGA_PORT_4_PIN_0
FPGA_PORT_4_PIN_1
FPGA_PORT_4_PIN_2
FPGA_PORT_4_PIN_3
FPGA_PORT_5_PIN_0
FPGA_PORT_5_PIN_1
FPGA_PORT_5_PIN_2
FPGA_PORT_5_PIN_3
FPGA_PORT_6_PIN_0
FPGA_PORT_6_PIN_1
FPGA_PORT_6_PIN_2
FPGA_PORT_6_PIN_3
FPGA_PORT_7_PIN_0
FPGA_PORT_7_PIN_1
FPGA_PORT_7_PIN_2
FPGA_PORT_7_PIN_3
FPGA_PORT_8_PIN_0
FPGA_PORT_8_PIN_1
FPGA_PORT_8_PIN_2
FPGA_PORT_8_PIN_3

File: fpga_mux.kicad_sch

Transceiver pair

FPGA_1_A
FPGA_1_B
FPGA_2_A
FPGA_2_B
1TX/RX+
1TX/RX−
2TX/RX+
2TX/RX−
PORT_1_DATA_1+
PORT_1_DATA_1−
PORT_1_DATA_2+
PORT_1_DATA_2−
CONFIG_DATA_IND
CONFIG_UPDATED
CONFIG_CLKD
CONFIG_OE_ND
CONFIG_CLR_ND
CONFIG_DATA_IN
CONFIG_UPDATE
CONFIG_CLK
CONFIG_OE_N
CONFIG_CLR_N
CONFIG_DATA_OUT

File: transceiver_pair.kicad_sch

Transceiver pair1

PORT_2_DATA_1+
PORT_2_DATA_1−
PORT_2_DATA_2+
PORT_2_DATA_2−

File: transceiver_pair.kicad_sch

Transceiver pair2

PORT_3_DATA_1+
PORT_3_DATA_1−
PORT_3_DATA_2+
PORT_3_DATA_2−

File: transceiver_pair.kicad_sch

Transceiver pair3

PORT_4_DATA_1+
PORT_4_DATA_1−
PORT_4_DATA_2+
PORT_4_DATA_2−

File: transceiver_pair.kicad_sch

Transceiver pair4

PORT_5_DATA_1+
PORT_5_DATA_1−
PORT_5_DATA_2+
PORT_5_DATA_2−

File: transceiver_pair.kicad_sch

Transceiver pair5

PORT_6_DATA_1+
PORT_6_DATA_1−
PORT_6_DATA_2+
PORT_6_DATA_2−

File: transceiver_pair.kicad_sch

Transceiver pair6

PORT_7_DATA_1+
PORT_7_DATA_1−
PORT_7_DATA_2+
PORT_7_DATA_2−

File: transceiver_pair.kicad_sch

Transceiver pair7

PORT_8_DATA_1+
PORT_8_DATA_1−
PORT_8_DATA_2+
PORT_8_DATA_2−

File: transceiver_pair.kicad_sch

Port Power

CONFIG_DATA_IN
CONFIG_UPDATE
CONFIG_CLKD
CONFIG_OE_ND
CONFIG_CLR_ND
PORT_1/2_VSENSE
PORT_3/4_VSENSE
PORT_5/6_VSENSE
PORT_7/8_VSENSE
5V_LOGIC_VSENSE
5V_PORT_VSENSE
LV_BANK_1_CURRENT_SENSE
LV_BANK_2_CURRENT_SENSE
HV_BANK_1_CURRENT_SENSE
HV_BANK_2_CURRENT_SENSE
LV_BANK_1_FAULT_N
LV_BANK_2_FAULT_N
HV_BANK_1_FAULT_N
HV_BANK_2_FAULT_N
PORT_1+
PORT_2+
PORT_3+
PORT_4+
PORT_5+
PORT_6+
PORT_7+
PORT_8+
CONFIG_DATA_OUT
CURRENT_SENSE_LV_BANK_1
CURRENT_SENSE_LV_BANK_2
CURRENT_SENSE_HV_BANK_1
CURRENT_SENSE_HV_BANK_2
FAULT_N_VL_BANK_1
FAULT_N_VL_BANK_2
FAULT_N_VH_BANK_1
FAULT_N_VH_BANK_2

File: port_power.kicad_sch

+3.3V
+5V_LOGIC
GND
C179 .1uf
C180 .1uf
C181 .1uf
C182 .1uf
C183 .1uf
C184 .1uf
C185 .1uf
C186 .1uf
C187 .1uf
C188 .1uf

U88 SN74LVC1T45DCK
U89 SN74LVC1T45DCK
U90 SN74LVC1T45DCK
U91 SN74LVC1T45DCK
U92 SN74LVC1T45DCK
VCCA
VCCB
A
B
DIR
GND
LED_SER_3V3
LED_SER
CONFIG_CLKD
LED_CLK
CONFIG_UPDATED
LED_LATCH
CONFIG_CLR_ND
LED_CLR_N
CONFIG_OE_ND
LED_OE_N

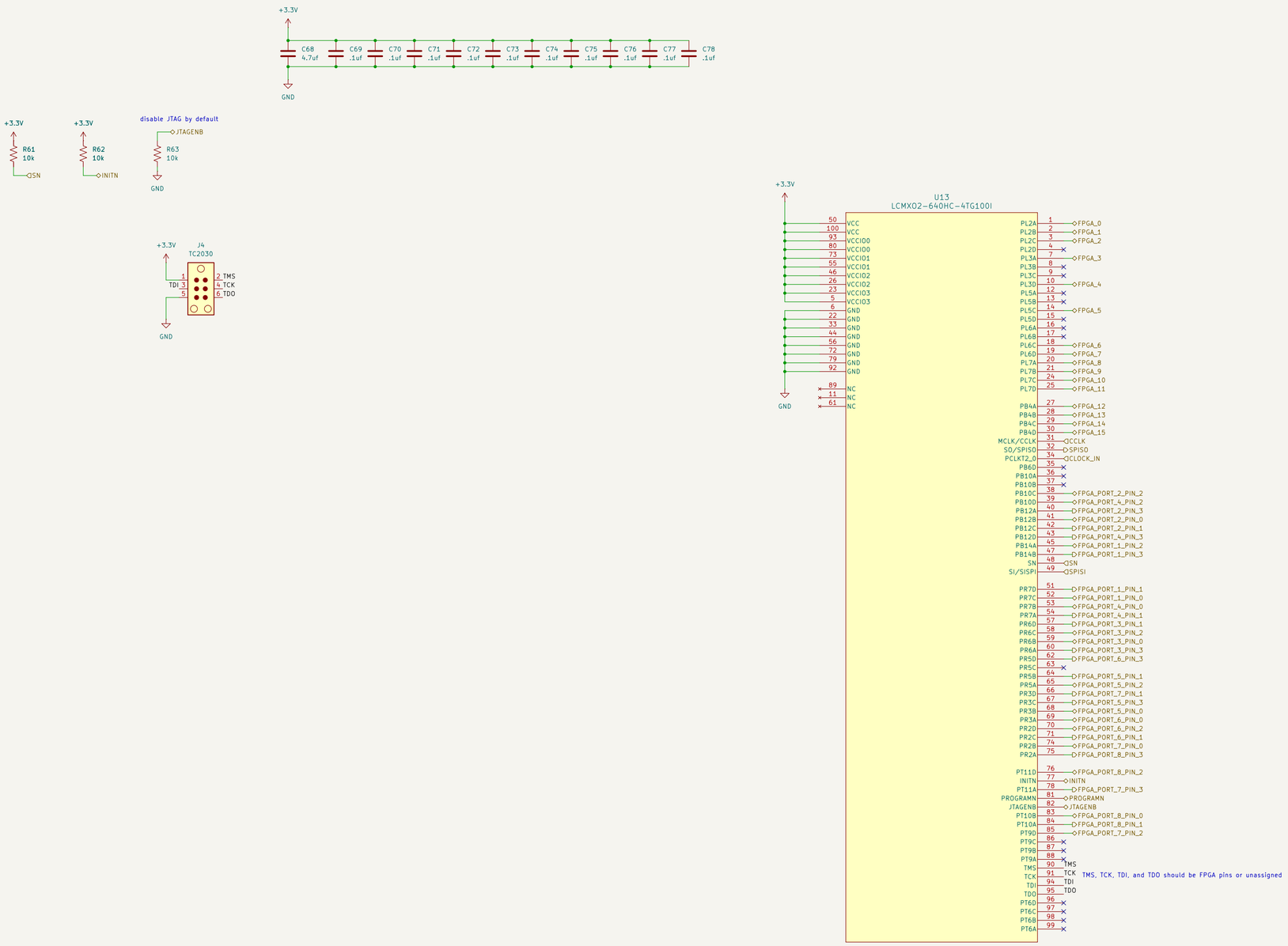
C101 .1uf
C102 .1uf
U52 74HC595
U53 74HC595
SER
SRCLK
SRCLR
RCLK
OE
QA
QB
QC
QD
QE
QF
QG
QH
VCC
LED_CLR_N
LED_LATCH
LED_OE_N
PORT_1_LED_GRN
PORT_1_LED_YLW
PORT_2_LED_GRN
PORT_2_LED_YLW
PORT_3_LED_GRN
PORT_3_LED_YLW
PORT_4_LED_GRN
PORT_4_LED_YLW
PORT_5_LED_GRN
PORT_5_LED_YLW
PORT_6_LED_GRN
PORT_6_LED_YLW
PORT_7_LED_GRN
PORT_7_LED_YLW
PORT_8_LED_GRN
PORT_8_LED_YLW

YELLOW
GREEN
SHIELD
R45 470 PORT_1_LED_YLW
R46 470 PORT_1_LED_GRN
R47 470 PORT_2_LED_YLW
R48 470 PORT_2_LED_GRN
R49 470 PORT_3_LED_YLW
R50 470 PORT_3_LED_GRN
R51 470 PORT_4_LED_YLW
R52 470 PORT_4_LED_GRN
R53 470 PORT_5_LED_YLW
R54 470 PORT_5_LED_GRN
R55 470 PORT_6_LED_YLW
R56 470 PORT_6_LED_GRN
R57 470 PORT_7_LED_YLW
R58 470 PORT_7_LED_GRN
R59 470 PORT_8_LED_YLW
R60 470 PORT_8_LED_GRN

Sheet: /Transceivers/
File: transceivers.kicad_sch

Title:

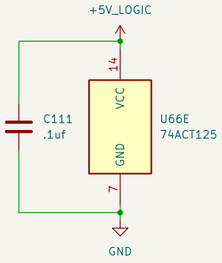
Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 4/20

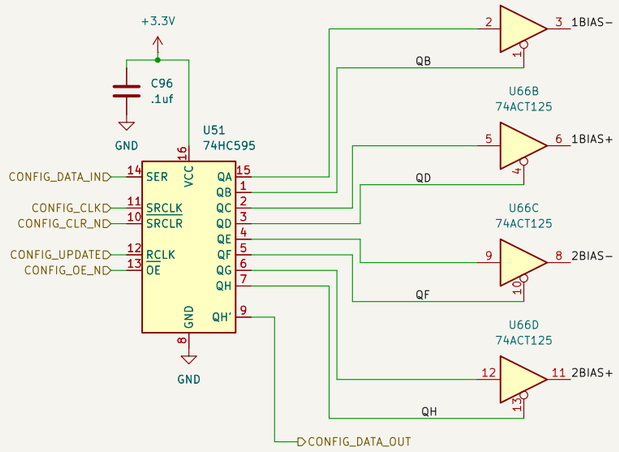


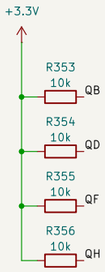
Sheet: /Transceivers/FPGA MUX/
File: fpga_mux.kicad_sch

Title:

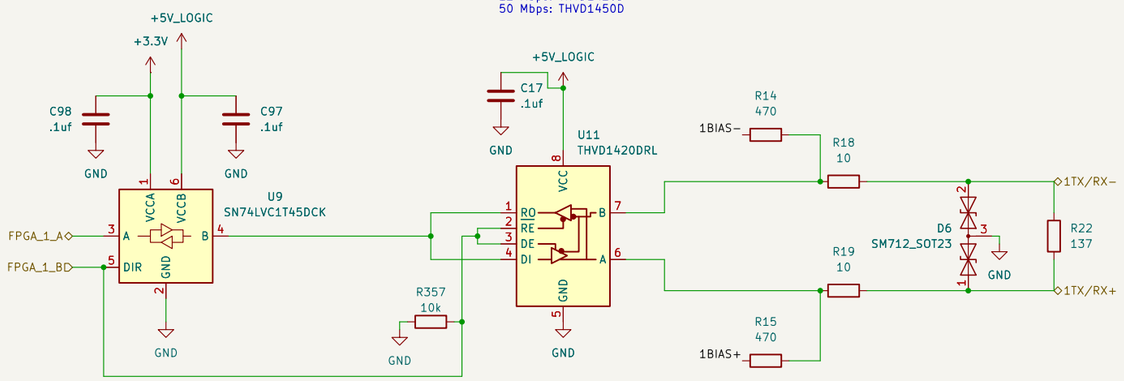
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 5/20

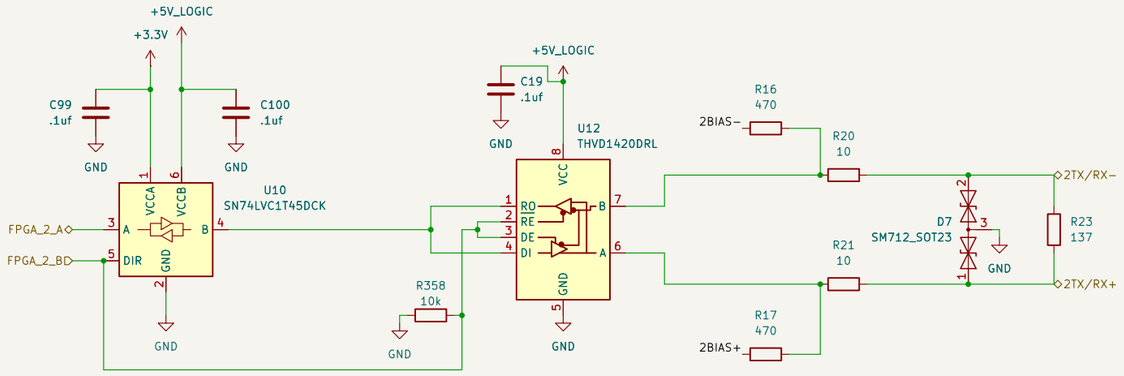












Sheet: /Transceivers/Transceiver pair/		
File: transceiver_pair.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 6/20

12 Mbps: THVD1420D
50 Mbps: THVD1450D

+5V_LOGIC
C117 .1uf
U67E 74ACT125
GND

+3.3V
C103 .1uf
GND
U54 74HC595
CONFIG_DATA_IND
CONFIG_CLKD
CONFIG_CLR_ND
CONFIG_UPDATED
CONFIG_OE_ND
SER
SRCLK
SRCLR
RCLK
OE
VCC
GND
QA
QB
QC
QD
QE
QF
QG
QH
QH'
CONFIG_DATA_OUT

U67A 74ACT125
U67B 74ACT125
U67C 74ACT125
U67D 74ACT125
1BIAS−
1BIAS+
2BIAS−
2BIAS+
QB
QD
QF
QH

+3.3V
R359 10k QB
R360 10k QD
R361 10k QF
R362 10k QH

+5V_LOGIC
C141 22uf
GND

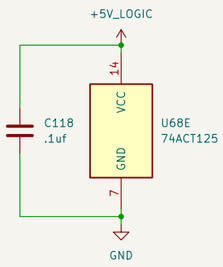
+5V_LOGIC
+3.3V
C104 .1uf
C106 .1uf
GND
U16 SN74LVC1T45DCK
VCCA
VCCB
FPGA_1_A
FPGA_1_BD
DIR
R363 10k

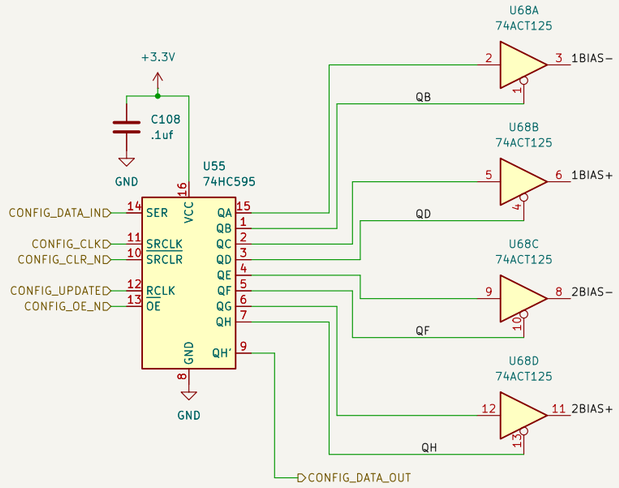
+5V_LOGIC
C25 .1uf
U26 THVD1420DRL
RO
RE
DE
DI
R64 470
R68 10
R69 10
R65 470
D36 SM712_SOT23
R72 137
1TX/RX−
1TX/RX+

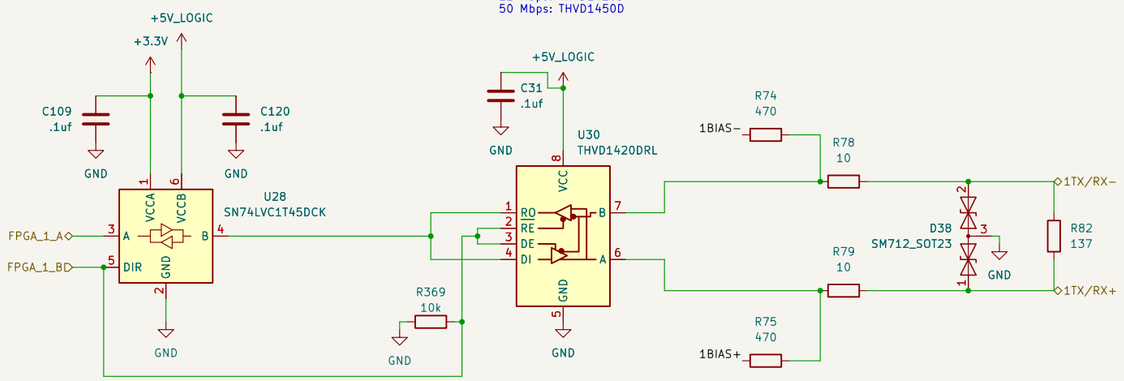
+5V_LOGIC
+3.3V
C105 .1uf
C107 .1uf
U17 SN74LVC1T45DCK
FPGA_2_A
FPGA_2_BD
R364 10k

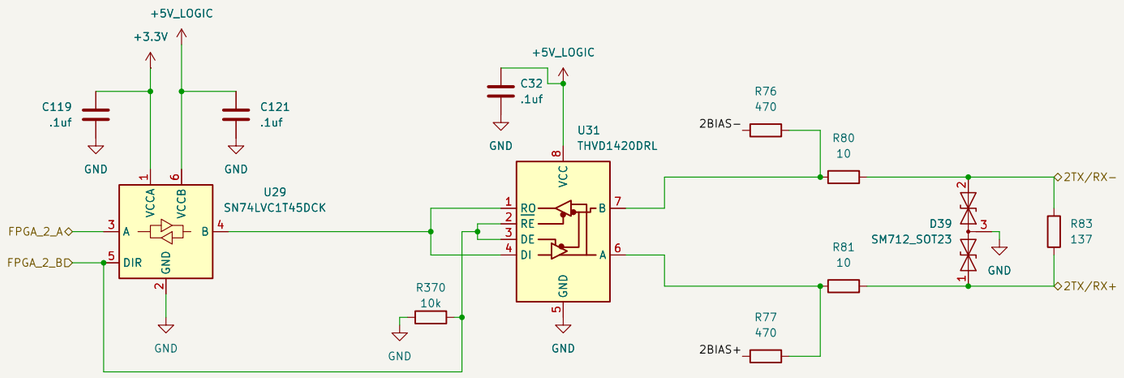
+5V_LOGIC
C26 .1uf
U27 THVD1420DRL
R66 470
R70 10
R71 10
R67 470
D37 SM712_SOT23
R73 137
2TX/RX−
2TX/RX+

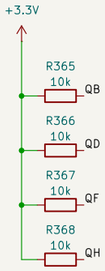
Sheet: /Transceivers/Transceiver pair1/
File: transceiver_pair.kicad_sch
Title:
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 7/20









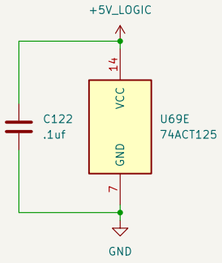


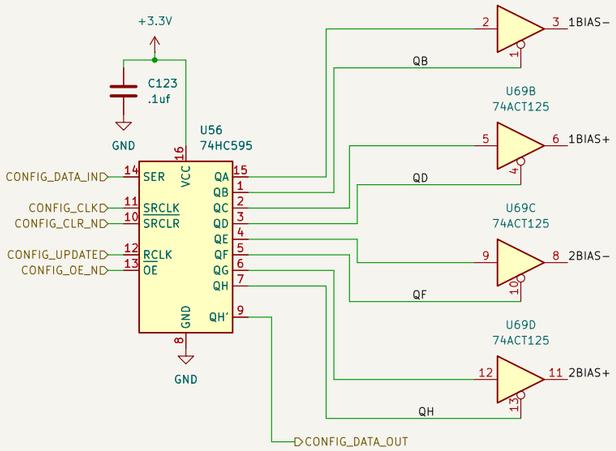


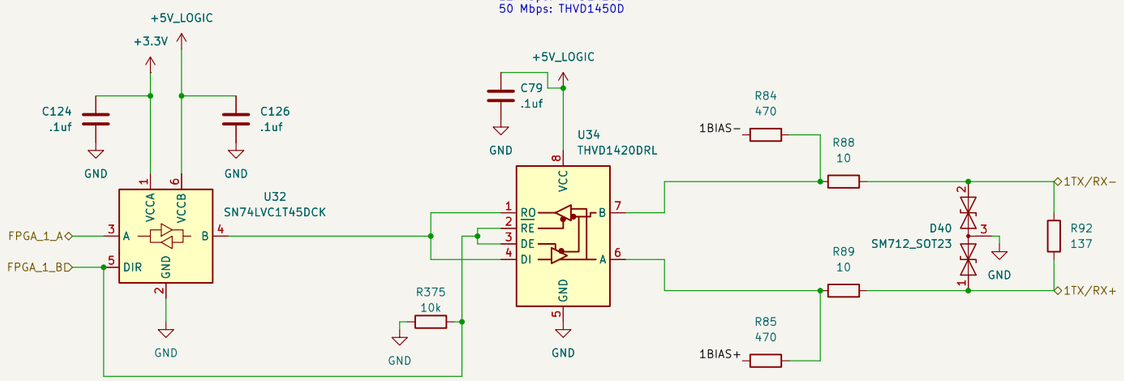
Sheet: /Transceivers/Transceiver pair2/
File: transceiver_pair.kicad_sch

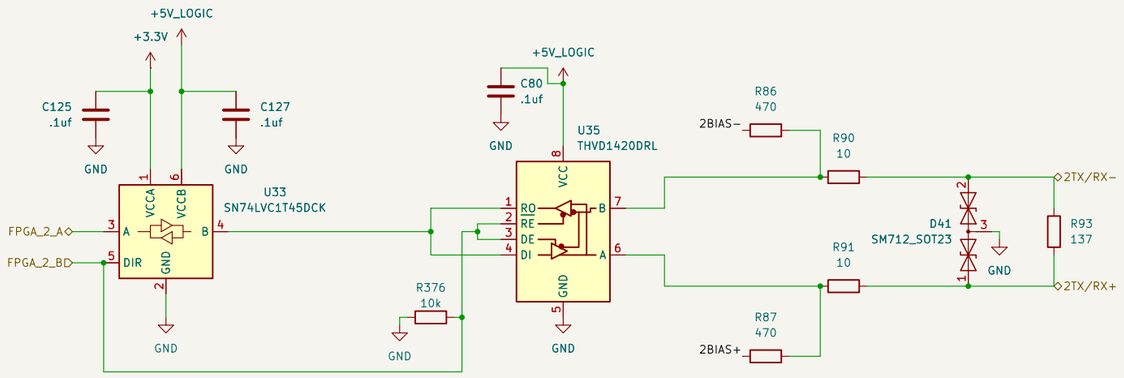
Title:

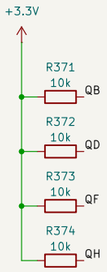
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 8/20









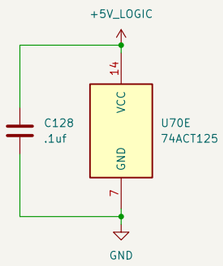


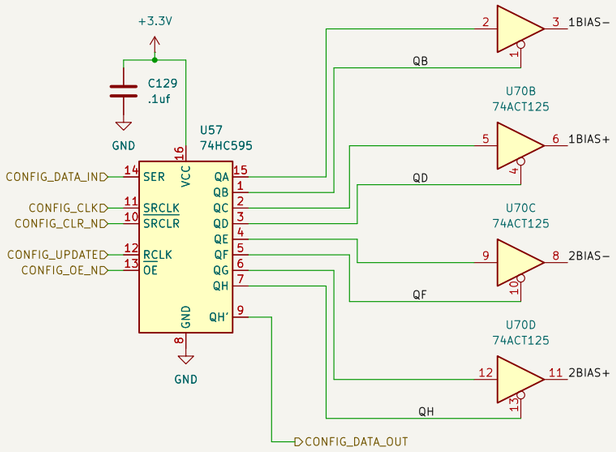


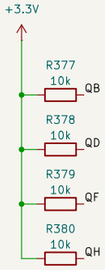
Sheet: /Transceivers/Transceiver pair3/
File: transceiver_pair.kicad_sch

Title:

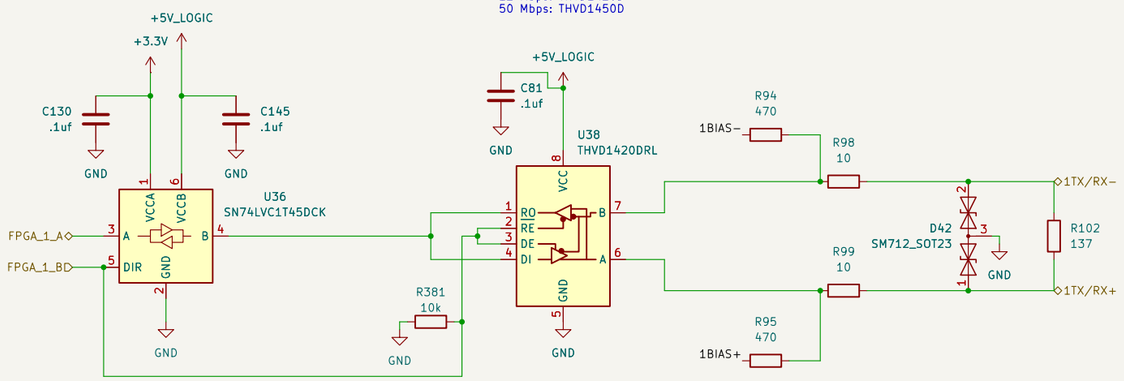
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 9/20

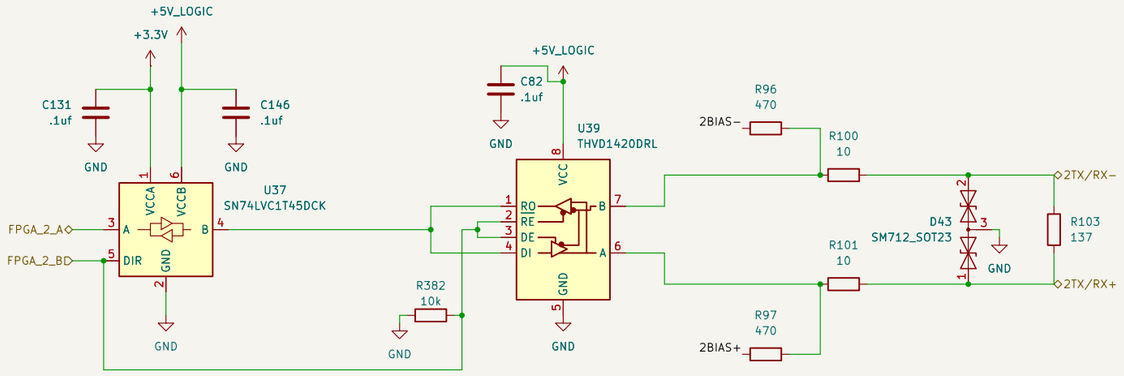












Sheet: /Transceivers/Transceiver pair4/		
File: transceiver_pair.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 10/20

12 Mbps: THVD1420D
50 Mbps: THVD1450D

+5V_LOGIC
C147
.1uf
U71E
74ACT125
GND

+3.3V
C160
.1uf
GND
U58
74HC595
CONFIG_DATA_IN
CONFIG_CLKD
CONFIG_CLR_ND
CONFIG_UPDATED
CONFIG_OE_ND
SER
SRCLK
SRCLR
RCLK
OE
VCC
GND
QA
QB
QC
QD
QE
QF
QG
QH
QH'
CONFIG_DATA_OUT
U71A
74ACT125
1BIAS−
U71B
74ACT125
1BIAS+
U71C
74ACT125
2BIAS−
U71D
74ACT125
2BIAS+

+3.3V
C161
.1uf
C163
.1uf
GND
U40
SN74LVC1T45DCK
VCCA
VCCB
A
B
DIR
FPGA_1_A
FPGA_1_BD
R407
10k
+5V_LOGIC
C83
.1uf
U42
THVD1420DRL
RO
RE
DE
DI
R104
470
R108
10
R109
10
R105
470
D44
SM712_SOT23
R112
137
1TX/RX−
1TX/RX+

+3.3V
C162
.1uf
C164
.1uf
U41
SN74LVC1T45DCK
FPGA_2_A
FPGA_2_BD
R408
10k
+5V_LOGIC
C84
.1uf
U43
THVD1420DRL
R106
470
R110
10
R111
10
R107
470
D45
SM712_SOT23
R113
137
2TX/RX−
2TX/RX+

+3.3V
R383
10k
QB
R384
10k
QD
R405
10k
QF
R406
10k
QH

+5V_LOGIC
C159
22uf
GND

Sheet: /Transceivers/Transceiver pair5/
File: transceiver_pair.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.8
Id: 11/20

12 Mbps: THVD1420D
50 Mbps: THVD1450D

+5V_LOGIC
C165 .1uf
U72E 74ACT125
GND

+3.3V
C167 .1uf
GND
U59 74HC595
CONFIG_DATA_IND
CONFIG_CLKD
CONFIG_CLR_ND
CONFIG_UPDATED
CONFIG_OE_ND
SER SRCLK SRCLR RCLK OE VCC GND QA QB QC QD QE QF QG QH QH'
CONFIG_DATA_OUT
U72A 74ACT125 1BIAS−
U72B 74ACT125 1BIAS+
U72C 74ACT125 2BIAS−
U72D 74ACT125 2BIAS+
QB QD QF QH

+3.3V
R409 10k QB
R410 10k QD
R411 10k QF
R412 10k QH

+5V_LOGIC
C166 22uf
GND

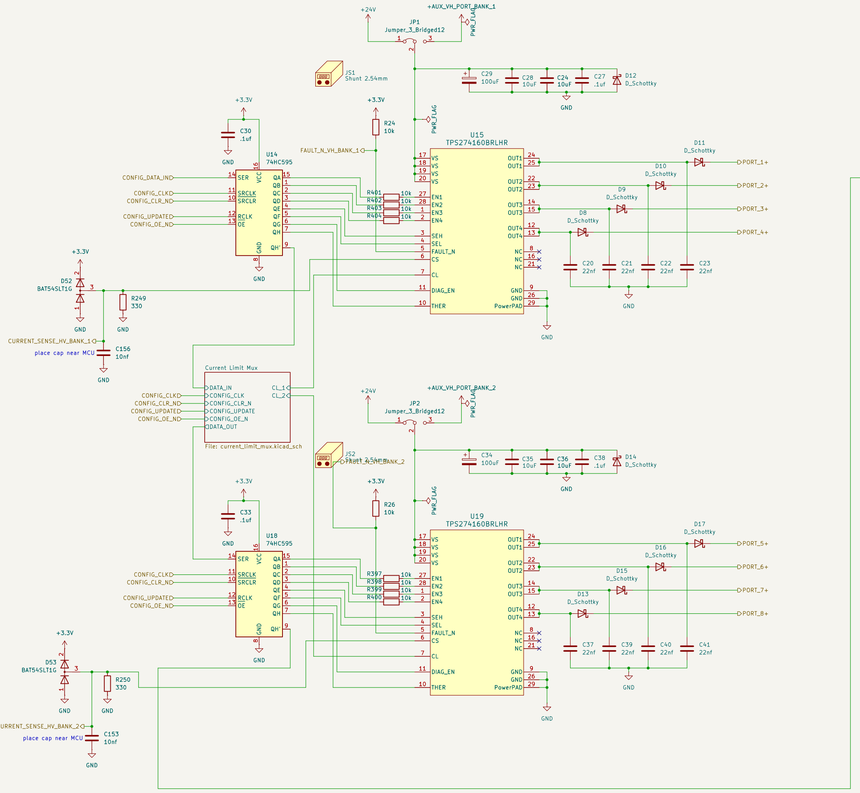
+5V_LOGIC
+3.3V
C168 .1uf
C170 .1uf
U44 SN74LVC1T45DCK
FPGA_1_A
FPGA_1_BD
VCCA VCCB A B DIR GND
R413 10k

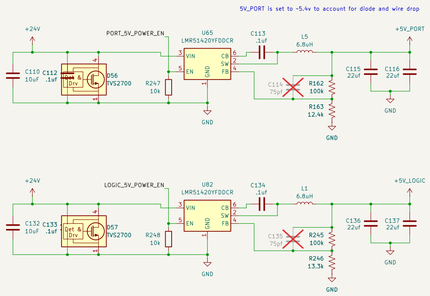
+5V_LOGIC
C85 .1uf
U46 THVD1420DRL
RO RE DE DI VCC GND A B
R114 470 1BIAS−
R118 10
R119 10
R115 470 1BIAS+
D46 SM712_SOT23
R122 137
1TX/RX−
1TX/RX+

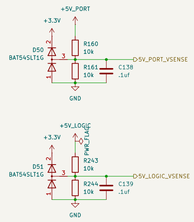
+5V_LOGIC
+3.3V
C169 .1uf
C171 .1uf
U45 SN74LVC1T45DCK
FPGA_2_A
FPGA_2_BD
R414 10k

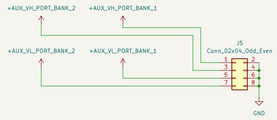
+5V_LOGIC
C86 .1uf
U47 THVD1420DRL
R116 470 2BIAS−
R120 10
R121 10
R117 470 2BIAS+
D47 SM712_SOT23
R123 137
2TX/RX−
2TX/RX+

Sheet: /Transceivers/Transceiver pair6/
File: transceiver_pair.kicad_sch
Title:
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 12/20





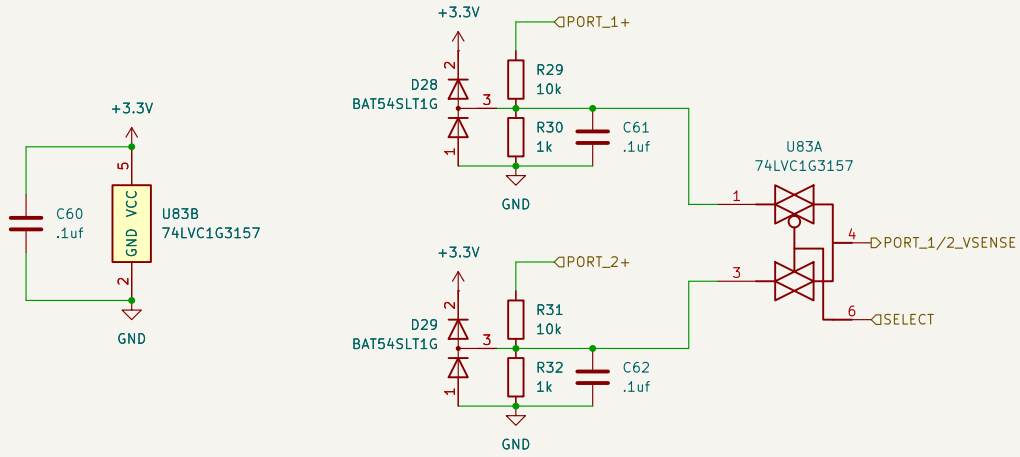




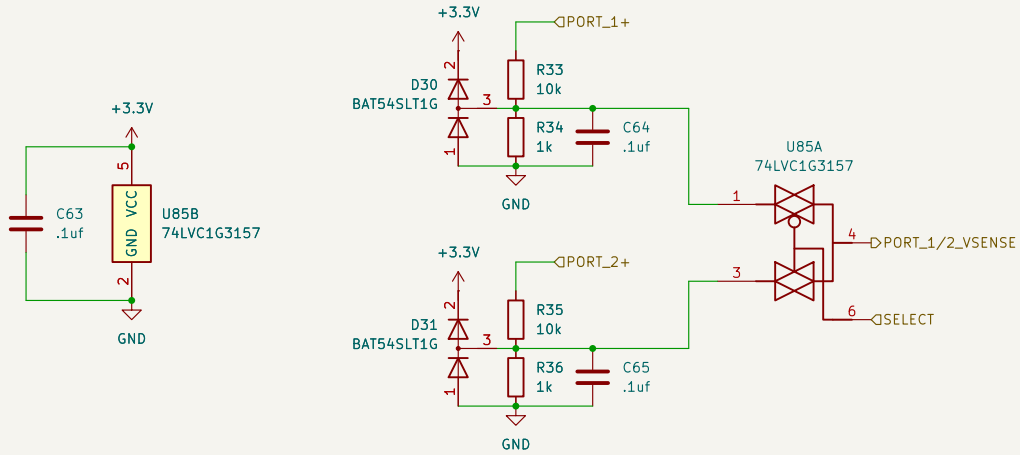
Sheet: /Transceivers/Port Power/
File: port_power.kicad_sch

Title:

Size: A1	Date:	Rev:
KiCad E.D.A. 10.0.8		Id: 13/20



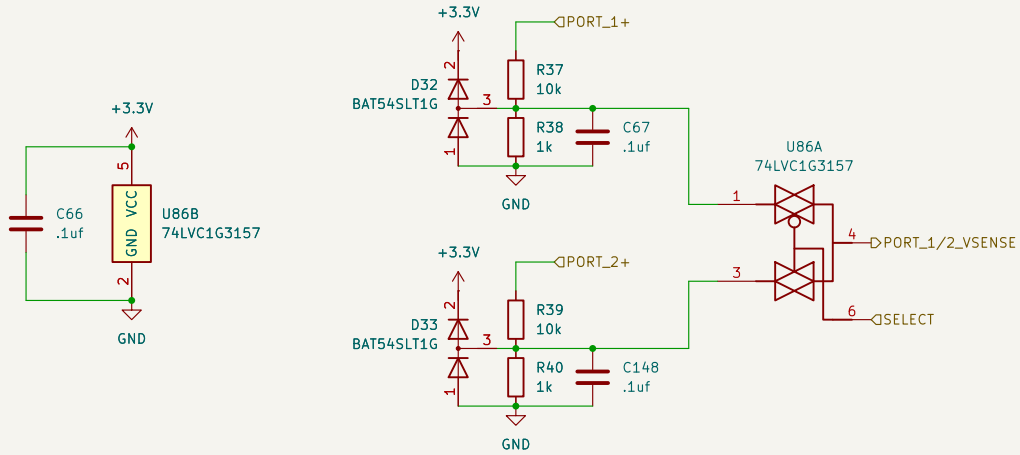
Sheet: /Transceivers/Port Power/Port Vsense Mux/		
File: port_vsense_mux.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 14/20



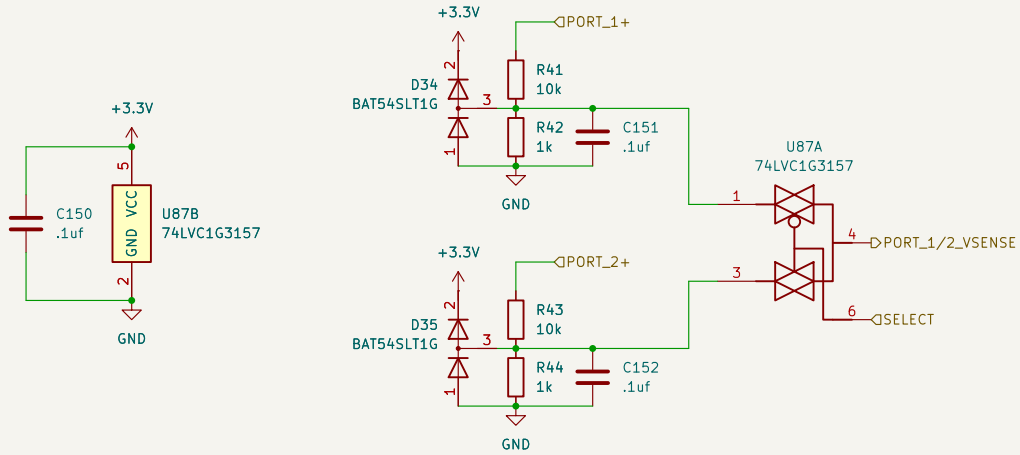
Sheet: /Transceivers/Port Power/Port Vsense Mux1/
File: port_vsense_mux.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 15/20



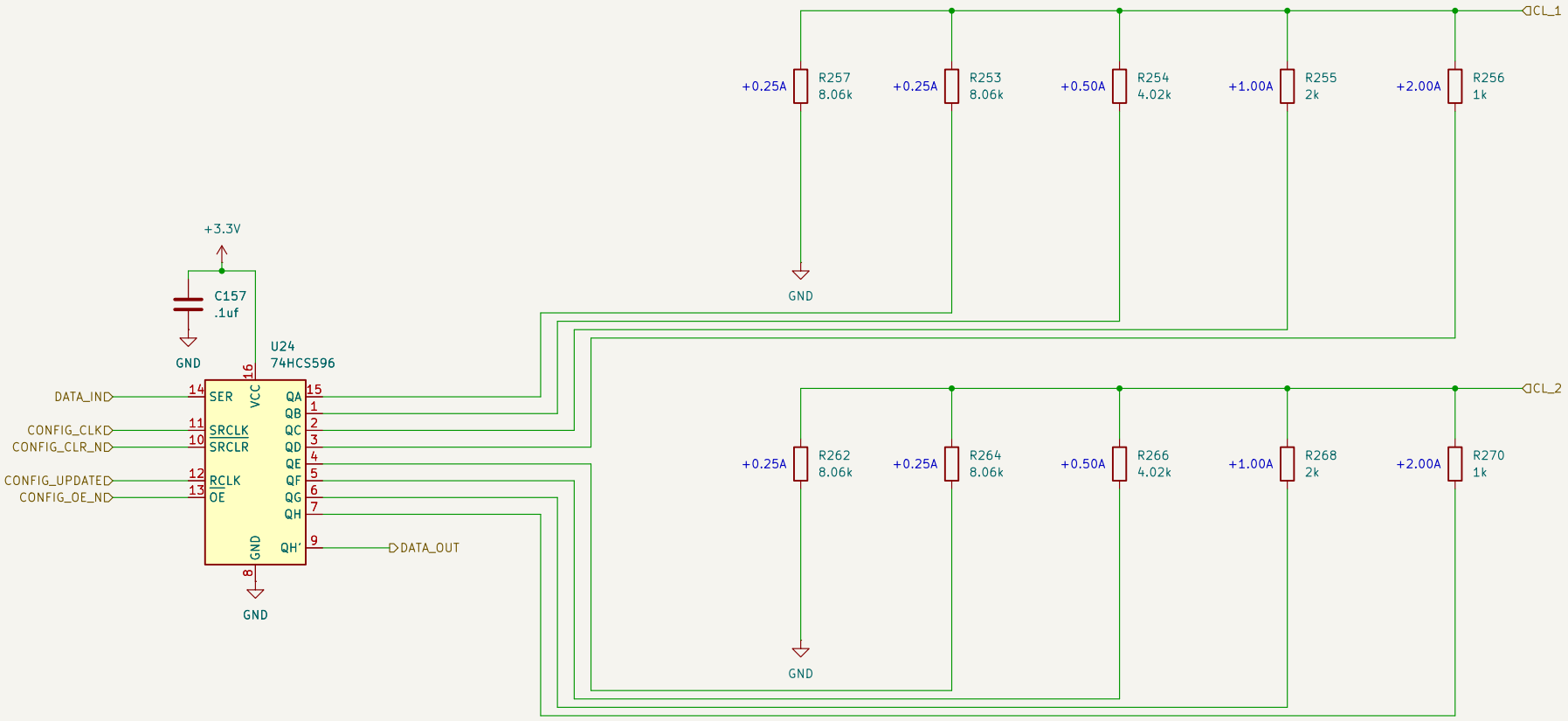
Sheet: /Transceivers/Port Power/Port Vsense Mux2/		
File: port_vsense_mux.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 16/20



Sheet: /Transceivers/Port Power/Port Vsense Mux3/
File: port_vsense_mux.kicad_sch

Title:

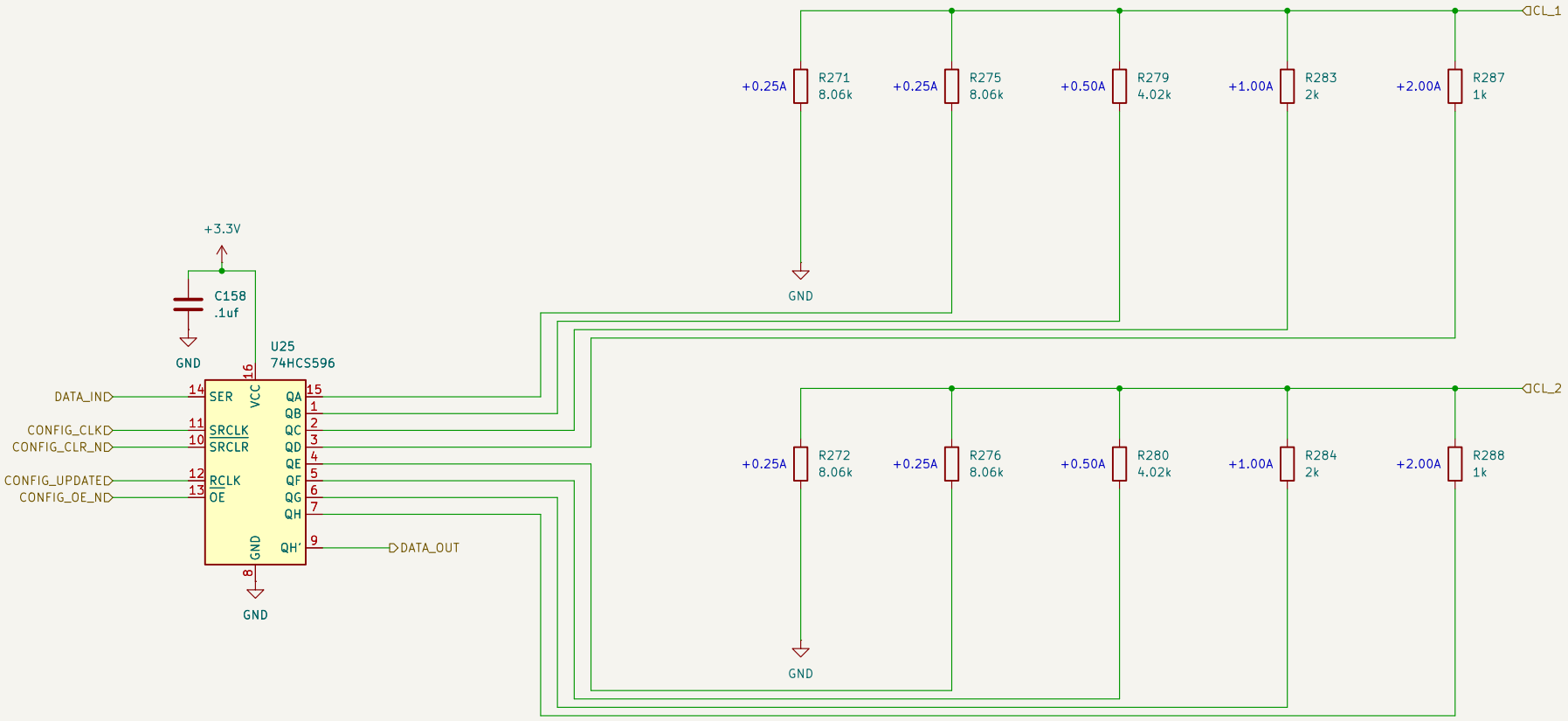
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 17/20



Sheet: /Transceivers/Port Power/Current Limit Mux/
File: current_limit_mux.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 18/20



Sheet: /Transceivers/Port Power/Current Limit Mux1/
File: current_limit_mux.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 19/20

12 Mbps: THVD1420D
50 Mbps: THVD1450D

+5V_LOGIC
C172 .1uf
U73E 74ACT125
GND

+3.3V
C174 .1uf
GND
U60 74HC595
CONFIG_DATA_IN
CONFIG_CLKD
CONFIG_CLR_ND
CONFIG_UPDATED
CONFIG_OE_ND
SER
SRCLK
SRCLR
RCLK
OE
QA
QB
QC
QD
QE
QF
QG
QH
QH'
CONFIG_DATA_OUT

U73A 74ACT125
U73B 74ACT125
U73C 74ACT125
U73D 74ACT125
1BIAS−
1BIAS+
2BIAS−
2BIAS+

+3.3V
C175 .1uf
C177 .1uf
U48 SN74LVC1T45DCK
FPGA_1_A
FPGA_1_BD
VCCA
VCCB
DIR
R419 10k

+5V_LOGIC
C87 .1uf
U50 THVD1420DRL
RO
RE
DE
DI
R124 470
R125 470
R128 10
R129 10
D48 SM712_SOT23
R132 137
1TX/RX−
1TX/RX+

+3.3V
C176 .1uf
C178 .1uf
U49 SN74LVC1T45DCK
FPGA_2_A
FPGA_2_BD
R420 10k

+5V_LOGIC
C88 .1uf
U61 THVD1420DRL
R126 470
R127 470
R130 10
R131 10
D49 SM712_SOT23
R133 137
2TX/RX−
2TX/RX+

+3.3V
R415 10k QB
R416 10k QD
R417 10k QF
R418 10k QH

+5V_LOGIC
C173 22uf
GND

Sheet: /Transceivers/Transceiver pair7/
File: transceiver_pair.kicad_sch
Title:
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 20/20